Supabase · PostgreSQL · HL7 FHIR R4

Integrated with Satu Sehat API — Kementerian Kesehatan RI

Version 1.0 · April 2026

Database Overview

This document describes the complete database schema for the SIMRS (Sistem Informasi Manajemen Rumah Sakit) EHR system built on Supabase (PostgreSQL). The schema is designed to support a comprehensive hospital management system with full integration to the Satu Sehat API from Indonesia's Ministry of Health (Kemenkes) using the HL7 FHIR R4 standard.

33 Tables	**~85 RLS Policies**	**8 Functions**	**20+ Triggers**	**14 Sections**

System Architecture

The database is organized into 14 logical sections. The **User Module** (publicly accessible by patients) covers appointments, queue management, and the check-in flow. The **System Module** (accessed by hospital staff) covers all clinical operations. All clinical data is mapped to FHIR R4 resources and synchronized to Satu Sehat using a dedicated sync log table.

Row Level Security (RLS)

Every table has RLS enabled. The security model follows these principles: (1) **Patients** can only read their own records across all clinical tables. (2) **Staff** can read all records within their organization. (3) **Write access is role-based**: only doctors write diagnoses, only pharmacists dispense medication, only lab nurses enter lab results, only nutritionists write nutrition orders. (4) **Admin** has broad access within their organization.

Satu Sehat Integration

Every table that maps to a FHIR resource contains: (a) an **ss_*_id** field to store the UUID returned by Satu Sehat after a successful POST, and (b) an **ss_sync_status** field with values pending | synced | failed | not_required. The **ss_sync_logs** table records every API call with full request/response payloads for debugging and compliance monitoring.

Automatic Triggers

Stock management: when a medication_dispense record is inserted, a trigger automatically deducts quantity from medication_stock. When a purchase_order_item's received quantity is updated, stock is automatically added. **Timestamps:** all major tables have an updated_at trigger. **Computed columns:** BMI (vital_signs), is_available (slots), is_below_minimum (stock), subtotal fields, and is_fully_received (PO items) are all computed and stored automatically.

Enumerated Types (ENUMs)

The following custom ENUM types are defined in the schema:

`user_role`	admin, doctor, nurse, lab_nurse, pharmacist, nutritionist, cashier, patient
`encounter_class`	outpatient, inpatient, emergency, observation
`encounter_status`	planned, arrived, in_progress, finished, cancelled
`appointment_status`	pending, booked, arrived, checked_in, cancelled, no_show
`payment_type`	umum, bpjs
`inpatient_status`	admitted, in_care, discharge_approved, discharged, bpjs_finalized
`emergency_status`	emergency_admitted, in_triage, in_treatment, completed, referred_out, admitted_to_inpatient
`surgery_status`	surgery_requested, surgery_scheduled, ready_for_surgery, intra_operative, surgery_completed, post_operative
`lab_order_status`	lab_ordered, sample_taken, processing, result_uploaded, verified
`po_status`	po_draft, po_sent, po_partially_received, po_completed, po_cancelled
`queue_status`	waiting, called, in_service, done, skipped
`gender`	male, female, other, unknown
`satu_sehat_sync_status`	pending, synced, failed, not_required

1. Master / Core Tables

Foundation tables that define the hospital structure, staff, and patients. All other modules depend on these tables.

organizations

FHIR: FHIR Organization

Purpose: Stores the hospital or clinic's profile. This is the top-level entity that all other data belongs to. Supports the CMS module where the owner can update clinic information.

Fields

id	UUID	Primary key	NO	PK
name	TEXT	Full name of the organization	NO	
type	TEXT	Organization type: puskesmas, klinik, rs	NO	
address	TEXT	Physical address	YES	
phone	TEXT	Contact phone number	YES	
email	TEXT	Contact email address	YES	
head_name	TEXT	Name of the clinic/hospital head	YES	
ss_organization_id	TEXT	Satu Sehat Organization ID (from Kemenkes)	YES	UNIQUE
operational_hours	JSONB	Operating hours per day stored as JSON object	YES	
logo_url	TEXT	URL to the organization logo in storage	YES	
is_active	BOOLEAN	Whether the organization is active	NO	DEFAULT TRUE
created_at	TIMESTAMPTZ	Record creation timestamp	NO	
updated_at	TIMESTAMPTZ	Last update timestamp (auto-managed)	NO	

locations

FHIR: FHIR Location

Purpose: Represents physical spaces inside the organization: poli rooms, IGD, operating rooms, wards, pharmacy, lab. Used in queue display, room assignment for inpatient, and surgery scheduling.

Fields

id	UUID	Primary key	NO	PK
organization_id	UUID	Parent organization	NO	FK → organizations
name	TEXT	Location name e.g. Poli Umum, IGD, Kamar 101	NO	
code	TEXT	Short code for the location	YES	
type	TEXT	Type: poli, ward, room, igd, ok, lab, pharmacy	NO	
floor	TEXT	Floor level	YES	

`capacity`	`INT`	Maximum capacity (beds/seats)	YES	DEFAULT 1
`ss_location_id`	`TEXT`	Satu Sehat Location ID	YES	UNIQUE
`is_active`	`BOOLEAN`	Whether the location is currently active	NO	DEFAULT TRUE
`created_at`	`TIMESTAMPTZ`	Record creation timestamp	NO	
`updated_at`	`TIMESTAMPTZ`	Last update timestamp	NO	

Foreign Key Relations

Column	References	Notes
`organization_id`	`organizations(id)`	CASCADE DELETE — removing org removes all locations

practitioners

FHIR: FHIR Practitioner

Purpose: Stores all hospital staff: doctors, nurses, lab nurses, pharmacists, nutritionists, cashiers, and admins. Each practitioner can be linked to a Supabase auth user for system login. Also sent to Satu Sehat as FHIR Practitioner.

Fields

`id`	`UUID`	Primary key	NO	PK
`user_id`	`UUID`	Linked Supabase auth user for login	YES	FK → auth.users, UNIQUE
`organization_id`	`UUID`	Organization this practitioner belongs to	NO	FK → organizations
`nik`	`TEXT`	National ID number (KTP)	YES	UNIQUE
`nip`	`TEXT`	Employee ID (Nomor Induk Pegawai)	YES	UNIQUE
`str_number`	`TEXT`	Surat Tanda Registrasi (doctors/nurses)	YES	
`sip_number`	`TEXT`	Surat Izin Praktik	YES	
`full_name`	`TEXT`	Full name of the practitioner	NO	
`role`	`user_role`	System role: admin, doctor, nurse, lab_nurse, pharmacist, nutritionist, cashier	NO	ENUM
`specialization`	`TEXT`	Medical specialization e.g. Dokter Umum	YES	
`gender`	`gender`	Gender enum: male, female, other, unknown	YES	ENUM
`phone`	`TEXT`	Contact phone	YES	
`email`	`TEXT`	Contact email	YES	
`ss_practitioner_id`	`TEXT`	Satu Sehat Practitioner ID	YES	UNIQUE
`is_active`	`BOOLEAN`	Active status	NO	DEFAULT TRUE
`created_at`	`TIMESTAMPTZ`	Record creation timestamp	NO	
`updated_at`	`TIMESTAMPTZ`	Last update timestamp	NO	

Foreign Key Relations

Column	References	Notes
`user_id`	`auth.users(id)`	SET NULL on delete — practitioner record preserved if auth account deleted
`organization_id`	`organizations(id)`	CASCADE DELETE

practitioner_roles

FHIR: FHIR PractitionerRole

Purpose: Assigns practitioners to specific poli/locations with a schedule. One doctor can serve in multiple poli on different days. Used in appointment slot generation and online booking.

Fields

`id`	`UUID`	Primary key	NO	PK
`practitioner_id`	`UUID`	The assigned practitioner	NO	FK → practitioners
`location_id`	`UUID`	The poli or room they are assigned to	NO	FK → locations
`schedule`	`JSONB`	Weekly schedule e.g. {days:[monday], start:08:00, end:14:00}	YES	
`is_active`	`BOOLEAN`	Active status	NO	DEFAULT TRUE
`ss_practitioner_role_id`	`TEXT`	Satu Sehat PractitionerRole ID	YES	UNIQUE
`created_at`	`TIMESTAMPTZ`	Record creation timestamp	NO	

Foreign Key Relations

Column	References	Notes
`practitioner_id`	`practitioners(id)`	CASCADE DELETE
`location_id`	`locations(id)`	CASCADE DELETE

patients

FHIR: FHIR Patient (MPI)

Purpose: Stores all patient records. NIK is validated against Kemenkes MPI (Master Patient Index) and the resulting Satu Sehat Patient ID is stored for all FHIR transactions. Patients can have a login account for the online booking portal.

Fields

`id`	`UUID`	Primary key	NO	PK
`user_id`	`UUID`	Optional: linked auth account for patient portal	YES	FK → auth.users, UNIQUE
`medical_record_no`	`TEXT`	Internal medical record number (auto-generated)	NO	UNIQUE
`nik`	`TEXT`	National ID — validated via Satu Sehat MPI	YES	UNIQUE
`bpjs_no`	`TEXT`	BPJS membership number	YES	UNIQUE
`full_name`	`TEXT`	Patient full name	NO	

date_of_birth	DATE	Date of birth	NO	
gender	gender	Gender enum	NO	ENUM
blood_type	TEXT	Blood type: A, B, AB, O	YES	
phone	TEXT	Contact phone	YES	
email	TEXT	Contact email	YES	
address	TEXT	Full address	YES	
emergency_contact_name	TEXT	Emergency contact full name	YES	
emergency_contact_phone	TEXT	Emergency contact phone	YES	
ss_patient_id	TEXT	Satu Sehat Patient ID from MPI lookup	YES	UNIQUE
ss_ihs_number	TEXT	IHS number from Satu Sehat	YES	UNIQUE
is_active	BOOLEAN	Patient active status	NO	DEFAULT TRUE
created_at	TIMESTAMPTZ	Record creation timestamp	NO	
updated_at	TIMESTAMPTZ	Last update timestamp	NO	

Foreign Key Relations

Column	References	Notes
user_id	auth.users(id)	SET NULL on delete

poli_services

FHIR: FHIR HealthcareService

Purpose: Defines the types of outpatient clinic services available (Poli Umum, Poli Gigi, Poli KIA, etc.). Each poli is linked to a location and can accept online bookings. The speciality_code is used for Satu Sehat HealthcareService registration.

Fields

id	UUID	Primary key	NO	PK
organization_id	UUID	Parent organization	NO	FK → organizations
location_id	UUID	Physical location of this poli	YES	FK → locations
name	TEXT	Service name e.g. Poli Umum	NO	
code	TEXT	Short code e.g. PU, PG	YES	UNIQUE
speciality_code	TEXT	Satu Sehat speciality code e.g. S001.09	YES	
quota_per_day	INT	Max patients per day	YES	DEFAULT 30
is_active	BOOLEAN	Active status	NO	DEFAULT TRUE
ss_healthcare_service_id	TEXT	Satu Sehat HealthcareService ID	YES	UNIQUE
created_at	TIMESTAMPTZ	Record creation timestamp	NO	

updated_at	TIMESTAMPTZ	Last update timestamp	NO	

Foreign Key Relations

Column	References	Notes
organization_id	organizations(id)	CASCADE DELETE
location_id	locations(id)	SET NULL on delete

2. Appointment & Online Booking

Handles the patient-facing online booking system (User Module). Patients can register, select a poli, choose a date and time slot, and receive a booking code for check-in.

appointment_slots

FHIR: FHIR Slot

Purpose: Represents available time slots for each poli and doctor. Used in the online booking calendar. The is_available computed column automatically reflects whether the slot is full.

Fields

id	UUID	Primary key	NO	PK
poli_service_id	UUID	Which poli this slot belongs to	NO	FK → poli_services
practitioner_id	UUID	Doctor for this slot	YES	FK → practitioners
slot_date	DATE	Date of the slot	NO	
start_time	TIME	Slot start time	NO	
end_time	TIME	Slot end time	NO	
quota	INT	Max patients for this slot	NO	DEFAULT 1
booked_count	INT	Current number of bookings	NO	DEFAULT 0
is_available	BOOLEAN	COMPUTED: TRUE if booked_count < quota	—	GENERATED STORED
ss_slot_id	TEXT	Satu Sehat Slot ID	YES	UNIQUE
created_at	TIMESTAMPTZ	Record creation timestamp	NO	

Foreign Key Relations

Column	References	Notes
poli_service_id	poli_services(id)	CASCADE DELETE
practitioner_id	practitioners(id)	SET NULL on delete

appointments

FHIR: FHIR Appointment

Purpose: Records a patient's booking for a specific poli visit. Contains BPJS referral info, payment type, and booking code. Sent to Satu Sehat as FHIR Appointment. The booking_code is used for check-in at the facility.

Fields

`id`	`UUID`	Primary key	NO	PK
`patient_id`	`UUID`	Patient making the booking	NO	FK → patients
`poli_service_id`	`UUID`	Target poli	NO	FK → poli_services
`practitioner_id`	`UUID`	Assigned doctor	YES	FK → practitioners
`slot_id`	`UUID`	Time slot selected	YES	FK → appointment_slots
`booking_code`	`TEXT`	Unique code shown to patient	NO	UNIQUE
`appointment_date`	`DATE`	Date of visit	NO	
`payment_type`	`payment_type`	umum or bpjs	NO	ENUM
`bpjs_referral_no`	`TEXT`	BPJS referral number if applicable	YES	
`chief_complaint`	`TEXT`	Patient's main complaint at booking	YES	
`status`	`appointment_status`	pending, booked, arrived, cancelled...	NO	ENUM
`ss_appointment_id`	`TEXT`	Satu Sehat Appointment ID	YES	UNIQUE
`ss_sync_status`	`ss_sync_status`	Sync state with Satu Sehat	NO	ENUM, DEFAULT pending
`created_at`	`TIMESTAMPTZ`	Record creation timestamp	NO	
`updated_at`	`TIMESTAMPTZ`	Last update timestamp	NO	

Foreign Key Relations

Column	References	Notes
`patient_id`	`patients(id)`	CASCADE DELETE
`poli_service_id`	`poli_services(id)`	
`slot_id`	`appointment_slots(id)`	

3. Queue System

Manages the physical queue at each poli. Queue numbers are auto-generated (e.g. A023). The queue display screen reads this table in real-time. Nurses call patients via the system which triggers audio announcements.

queues

FHIR: — (internal only)

Purpose: Tracks the daily queue for each poli. Each row represents one patient's place in line. The queue display (User Module) reads current and upcoming queue numbers from this table. Nurses update the status to move the queue forward.

Fields

`id`	`UUID`	Primary key	NO	PK
`appointment_id`	`UUID`	Linked appointment if booked online	YES	FK → appointments
`patient_id`	`UUID`	Patient in queue	NO	FK → patients
`poli_service_id`	`UUID`	Which poli queue	NO	FK → poli_services
`location_id`	`UUID`	Physical location/room	YES	FK → locations
`queue_date`	`DATE`	Date of this queue entry	NO	DEFAULT CURRENT_DATE
`queue_number`	`TEXT`	Display number e.g. A023	NO	
`queue_prefix`	`TEXT`	Letter prefix e.g. A, B	NO	DEFAULT 'A'
`sequence_number`	`INT`	Numeric sequence for ordering	NO	
`status`	`queue_status`	waiting, called, in_service, done, skipped	NO	ENUM
`called_at`	`TIMESTAMPTZ`	When the patient was called	YES	
`served_at`	`TIMESTAMPTZ`	When service started	YES	
`done_at`	`TIMESTAMPTZ`	When service completed	YES	
`called_by`	`UUID`	Nurse who called this patient	YES	FK → practitioners
`created_at`	`TIMESTAMPTZ`	Record creation timestamp	NO	

Foreign Key Relations

Column	References	Notes
`appointment_id`	`appointments(id)`	
`patient_id`	`patients(id)`	
`poli_service_id`	`poli_services(id)`	
`called_by`	`practitioners(id)`	

4. Encounters

The central clinical entity. Every patient-provider interaction — outpatient visit, inpatient stay, emergency visit, or referral — is an Encounter. All clinical records (vital signs, diagnoses, prescriptions, etc.) are linked to an Encounter.

episodes_of_care

FHIR: FHIR EpisodeOfCare

Purpose: Groups all encounters during one inpatient stay. Created when a patient is admitted and closed on discharge. The running bill accumulates charges under this episode. Sent to Satu Sehat as FHIR EpisodeOfCare.

Fields

`id`	`UUID`	Primary key	NO	PK
`patient_id`	`UUID`	Admitted patient	NO	FK → patients
`organization_id`	`UUID`	Admitting organization	NO	FK → organizations
`start_date`	`DATE`	Admission date	NO	
`end_date`	`DATE`	Discharge date	YES	
`status`	`inpatient_status`	admitted → in_care → discharged...	NO	ENUM
`diagnosis_primary`	`TEXT`	Primary ICD-10 diagnosis code	YES	
`room_location_id`	`UUID`	Room/ward assigned	YES	FK → locations
`bed_number`	`TEXT`	Assigned bed number	YES	
`dpjp_id`	`UUID`	Dokter Penanggung Jawab Pasien	YES	FK → practitioners
`ss_episode_of_care_id`	`TEXT`	Satu Sehat EpisodeOfCare ID	YES	UNIQUE
`ss_sync_status`	`ss_sync_status`	Sync state with Satu Sehat	NO	ENUM
`created_at`	`TIMESTAMPTZ`	Record creation timestamp	NO	
`updated_at`	`TIMESTAMPTZ`	Last update timestamp	NO	

Foreign Key Relations

Column	References	Notes
`patient_id`	`patients(id)`	
`organization_id`	`organizations(id)`	
`room_location_id`	`locations(id)`	
`dpjp_id`	`practitioners(id)`	

encounters

FHIR: FHIR Encounter

Purpose: The core table of the entire system. Every clinical visit — outpatient check-in, daily inpatient round, IGD admission, surgery — creates one row here. All clinical sub-records (diagnoses, vitals, prescriptions) reference this table.

Fields

`id`	`UUID`	Primary key	NO	PK
`patient_id`	`UUID`	The patient being seen	NO	FK → patients

`organization_id`	`UUID`	Facility where encounter occurs	NO	FK → organizations
`location_id`	`UUID`	Specific room/poli	YES	FK → locations
`poli_service_id`	`UUID`	Poli service for outpatient visits	YES	FK → poli_services
`appointment_id`	`UUID`	Source appointment if pre-booked	YES	FK → appointments
`episode_of_care_id`	`UUID`	Parent episode (inpatient only)	YES	FK → episodes_of_care
`queue_id`	`UUID`	Queue entry for this visit	YES	FK → queues
`doctor_id`	`UUID`	Attending doctor	YES	FK → practitioners
`nurse_id`	`UUID`	Attending nurse	YES	FK → practitioners
`encounter_class`	`encounter_class`	outpatient, inpatient, emergency	NO	ENUM
`status`	`encounter_status`	planned, arrived, in_progress, finished	NO	ENUM
`payment_type`	`payment_type`	umum or bpjs	NO	ENUM
`arrived_at`	`TIMESTAMPTZ`	When patient physically arrived	YES	
`started_at`	`TIMESTAMPTZ`	When care began	YES	
`finished_at`	`TIMESTAMPTZ`	When encounter ended	YES	
`is_referred_in`	`BOOLEAN`	True if patient came via referral letter	NO	DEFAULT FALSE
`ss_encounter_id`	`TEXT`	Satu Sehat Encounter ID — store this!	YES	UNIQUE
`ss_sync_status`	`ss_sync_status`	Sync state with Satu Sehat	NO	ENUM
`created_at`	`TIMESTAMPTZ`	Record creation timestamp	NO	
`updated_at`	`TIMESTAMPTZ`	Last update timestamp	NO	

Foreign Key Relations

Column	References	Notes
`patient_id`	`patients(id)`	
`organization_id`	`organizations(id)`	
`appointment_id`	`appointments(id)`	
`episode_of_care_id`	`episodes_of_care(id)`	
`doctor_id`	`practitioners(id)`	
`nurse_id`	`practitioners(id)`	

5. Clinical Records

All clinical data captured during an encounter. These tables collectively form the Electronic Medical Record (EMR/RME). Each table maps to a specific FHIR resource and is synced to Satu Sehat.

vital_signs

FHIR: FHIR Observation

Purpose: Recorded by nurses when a patient's queue number is called. Includes blood pressure, temperature, weight, height, and BMI (auto-calculated). Sent to Satu Sehat as FHIR Observation.

Fields

`id`	`UUID`	Primary key	NO	PK
`encounter_id`	`UUID`	Parent encounter	NO	FK → encounters
`patient_id`	`UUID`	Patient	NO	FK → patients
`recorded_by`	`UUID`	Nurse who recorded	NO	FK → practitioners
`recorded_at`	`TIMESTAMPTZ`	Recording timestamp	NO	
`systolic_bp`	`INT`	Systolic blood pressure (mmHg)	YES	
`diastolic_bp`	`INT`	Diastolic blood pressure (mmHg)	YES	
`heart_rate`	`INT`	Heart rate (bpm)	YES	
`temperature`	`DECIMAL(4,1)`	Body temperature (Celsius)	YES	
`oxygen_saturation`	`DECIMAL(5,2)`	SpO2 (%)	YES	
`weight_kg`	`DECIMAL(5,2)`	Weight in kilograms	YES	
`height_cm`	`DECIMAL(5,2)`	Height in centimeters	YES	
`bmi`	`DECIMAL(5,2)`	COMPUTED: weight/(height/100)^2	—	GENERATED STORED
`pain_scale`	`INT`	Pain scale 0–10	YES	CHECK 0–10
`ss_observation_id`	`TEXT`	Satu Sehat Observation ID	YES	

Foreign Key Relations

Column	References	Notes
`encounter_id`	`encounters(id)`	CASCADE DELETE
`recorded_by`	`practitioners(id)`	

diagnoses

FHIR: FHIR Condition

Purpose: ICD-10 coded diagnoses entered by doctors. Each encounter can have one primary and multiple secondary diagnoses. Sent to Satu Sehat as FHIR Condition.

Fields

`id`	`UUID`	Primary key	NO	PK
`encounter_id`	`UUID`	Parent encounter	NO	FK → encounters
`patient_id`	`UUID`	Patient	NO	FK → patients
`recorded_by`	`UUID`	Doctor who recorded	NO	FK → practitioners
`icd10_code`	`TEXT`	ICD-10 diagnosis code e.g. J06.9	NO	

icd10_display	TEXT	Human-readable diagnosis name	NO	
diagnosis_type	TEXT	primary, secondary, or comorbidity	NO	DEFAULT primary
clinical_status	TEXT	active, resolved, inactive	NO	DEFAULT active
onset_date	DATE	When the condition started	YES	
ss_condition_id	TEXT	Satu Sehat Condition ID	YES	UNIQUE

Foreign Key Relations

Column	References	Notes
encounter_id	encounters(id)	CASCADE DELETE
recorded_by	practitioners(id)	

procedures

FHIR: FHIR Procedure

Purpose: Medical procedures and actions performed during an encounter. Uses ICD-9-CM or KPTL procedure codes. Sent to Satu Sehat as FHIR Procedure. Surgery is flagged with is_surgery = TRUE.

Fields

id	UUID	Primary key	NO	PK
encounter_id	UUID	Parent encounter	NO	FK → encounters
patient_id	UUID	Patient	NO	FK → patients
performed_by	UUID	Practitioner who performed	NO	FK → practitioners
procedure_code	TEXT	ICD-9-CM or KPTL code	NO	
procedure_display	TEXT	Human-readable procedure name	NO	
performed_at	TIMESTAMPTZ	Timestamp of procedure	NO	
is_surgery	BOOLEAN	True if this is a surgical procedure	NO	DEFAULT FALSE
ss_procedure_id	TEXT	Satu Sehat Procedure ID	YES	UNIQUE

Foreign Key Relations

Column	References	Notes
encounter_id	encounters(id)	CASCADE DELETE
performed_by	practitioners(id)	

clinical_notes

FHIR: FHIR ClinicalImpression

Purpose: CPPT (Catatan Perkembangan Pasien Terintegrasi) — the integrated clinical progress notes. All clinical staff (doctors, nurses, nutritionists, lab nurses) write SOAP-format notes here during each encounter. Sent to Satu Sehat as FHIR ClinicalImpression.

Fields

id	UUID	Primary key	NO	PK
encounter_id	UUID	Parent encounter	NO	FK → encounters
patient_id	UUID	Patient	NO	FK → patients
written_by	UUID	Author (any clinical staff)	NO	FK → practitioners
writer_role	user_role	Role at time of writing	NO	ENUM
note_date	TIMESTAMPTZ	Date and time of note	NO	
subjective	TEXT	S: Patient's reported complaints	YES	
objective	TEXT	O: Clinical examination findings	YES	
assessment	TEXT	A: Clinical assessment	YES	
plan	TEXT	P: Plan and follow-up actions	YES	
ss_clinical_impression_id	TEXT	Satu Sehat ClinicalImpression ID	YES	UNIQUE

Foreign Key Relations

Column	References	Notes
encounter_id	encounters(id)	CASCADE DELETE
written_by	practitioners(id)	

allergy_intolerances

FHIR: FHIR AllergyIntolerance

Purpose: Patient allergy records. Persists across encounters as part of the patient's permanent medical history. Can be created or updated at any encounter. Sent to Satu Sehat as FHIR AllergyIntolerance.

Fields

id	UUID	Primary key	NO	PK
patient_id	UUID	Patient with the allergy	NO	FK → patients
recorded_by	UUID	Practitioner who documented it	NO	FK → practitioners
encounter_id	UUID	Encounter where it was first noted	YES	FK → encounters
substance_display	TEXT	Name of the allergen	NO	
category	TEXT	medication, food, environment	NO	DEFAULT medication
criticality	TEXT	low, high, unable-to-assess	NO	
reaction_description	TEXT	Description of allergic reaction	YES	
is_active	BOOLEAN	Currently active allergy	NO	DEFAULT TRUE
ss_allergy_id	TEXT	Satu Sehat AllergyIntolerance ID	YES	UNIQUE

Foreign Key Relations

Column	References	Notes
patient_id	patients(id)	CASCADE DELETE
recorded_by	practitioners(id)	
encounter_id	encounters(id)	

informed_consents

FHIR: — (internal, referenced in Procedure)

Purpose: Records patient or guardian consent for invasive procedures and surgeries. Includes who consented, their relation to the patient, and a link to the signed document stored in Supabase Storage.

Fields

id	UUID	Primary key	NO	PK
encounter_id	UUID	Parent encounter	NO	FK → encounters
patient_id	UUID	Patient	NO	FK → patients
procedure_id	UUID	Related procedure	YES	FK → procedures
consent_type	TEXT	surgery, invasive_procedure, general	NO	
consented_by	TEXT	Name of person who consented	NO	
relation_to_patient	TEXT	self, parent, spouse, etc.	YES	
document_url	TEXT	URL to signed consent document	YES	
consented_at	TIMESTAMPTZ	Timestamp of consent	NO	

Foreign Key Relations

Column	References	Notes
encounter_id	encounters(id)	CASCADE DELETE
procedure_id	procedures(id)	

medical_resumes

FHIR: FHIR Composition

Purpose: The final medical summary (Resume Medis) created by the doctor at the end of an outpatient or inpatient encounter. This is the key document for Satu Sehat compliance, sent as FHIR Composition (Resume Medis Rawat Jalan / Rawat Inap).

Fields

id	UUID	Primary key	NO	PK
encounter_id	UUID	Parent encounter	NO	FK → encounters
patient_id	UUID	Patient	NO	FK → patients
authored_by	UUID	Doctor who authored	NO	FK → practitioners
chief_complaint	TEXT	Main complaint	YES	

history_of_illness	TEXT	History of present illness	YES	
physical_examination	TEXT	Physical exam findings	YES	
summary	TEXT	Overall clinical summary	YES	
follow_up_plan	TEXT	Post-visit instructions/plan	YES	
ss_composition_id	TEXT	Satu Sehat Composition ID	YES	UNIQUE
ss_sync_status	ss_sync_status	Sync state	NO	ENUM

Foreign Key Relations

Column	References	Notes
encounter_id	encounters(id)	CASCADE DELETE
authored_by	practitioners(id)	

6. Emergency (IGD)

Extends the base Encounter with IGD-specific data: triage information, triage category (P1–P4), resuscitation notes, and final outcome (discharged, referred, or admitted to inpatient).

emergency_encounters

FHIR: FHIR Encounter (class=emergency)

Purpose: Extends the base encounters table with IGD-specific fields. Created when a patient arrives at IGD. Triage is performed by the IGD nurse. Outcome determines whether the patient is discharged, referred to another facility, or converted to an inpatient admission.

Fields

id	UUID	Primary key	NO	PK
encounter_id	UUID	Base encounter (1-to-1)	NO	FK → encounters, UNIQUE
patient_id	UUID	Patient	NO	FK → patients
status	emergency_status	emergency_admitted → completed...	NO	ENUM
triage_category	TEXT	P1 (merah), P2, P3, P4 (hitam)	YES	
triage_complaint	TEXT	Chief complaint at triage	YES	
triage_nurse_id	UUID	Nurse who performed triage	YES	FK → practitioners
triaged_at	TIMESTAMPTZ	Triage timestamp	YES	
is_critical	BOOLEAN	True if immediate resuscitation needed	NO	DEFAULT FALSE
resuscitation_notes	TEXT	Notes from resuscitation actions	YES	
outcome	TEXT	discharged, referred, admitted_inpatient	YES	
referred_to	TEXT	Destination facility if referred	YES	

needs_ambulance	BOOLEAN	Whether ambulance was requested	NO	DEFAULT FALSE

Foreign Key Relations

Column	References	Notes
encounter_id	encounters(id)	CASCADE DELETE — 1:1 with encounters
triage_nurse_id	practitioners(id)	

7. Inpatient (Rawat Inap)

Manages the full inpatient stay lifecycle from admission to discharge. Includes daily care records, running bill accumulation, DPJP assignment, and BPJS claim finalization.

inpatient_admissions

FHIR: — (part of FHIR EpisodeOfCare / Encounter)

Purpose: Detailed record of a patient's hospital admission. Tracks room assignment, DPJP (Dokter Penanggung Jawab Pasien), admission source, discharge approval, and BPJS SEP number. Linked to an EpisodeOfCare which groups all encounters during the stay.

Fields

id	UUID	Primary key	NO	PK
episode_of_care_id	UUID	Parent episode (1-to-1)	NO	FK → episodes_of_care, UNIQUE
patient_id	UUID	Admitted patient	NO	FK → patients
admitted_from	TEXT	outpatient, igd, rujukan	NO	DEFAULT outpatient
admission_date	TIMESTAMPTZ	Date and time of admission	NO	
discharge_date	TIMESTAMPTZ	Date and time of discharge	YES	
dpjp_id	UUID	Primary attending physician	NO	FK → practitioners
room_location_id	UUID	Assigned room/ward	NO	FK → locations
bed_number	TEXT	Bed identifier	NO	
room_class	TEXT	kelas_1, kelas_2, kelas_3, vip	NO	
status	inpatient_status	admitted → in_care → discharged	NO	ENUM
discharge_approved_by	UUID	Doctor who approved discharge	YES	FK → practitioners
discharge_summary	TEXT	Discharge summary notes	YES	
sep_number	TEXT	BPJS SEP number	YES	

Foreign Key Relations

Column	References	Notes
`episode_of_care_id`	`episodes_of_care(id)`	CASCADE DELETE
`dpjp_id`	`practitioners(id)`	
`room_location_id`	`locations(id)`	

inpatient_daily_records

FHIR: — (linked via encounters)

Purpose: Links a daily inpatient encounter to the admission. Each day (and shift) creates one encounter and one daily record. This provides a structured timeline of the patient's hospital stay.

Fields

`id`	`UUID`	Primary key	NO	PK
`admission_id`	`UUID`	Parent admission	NO	FK → inpatient_admissions
`encounter_id`	`UUID`	The encounter for this day	NO	FK → encounters
`record_date`	`DATE`	Date of this record	NO	DEFAULT CURRENT_DATE
`shift`	`TEXT`	pagi, sore, malam	YES	DEFAULT pagi
`created_at`	`TIMESTAMPTZ`	Record creation timestamp	NO	

Foreign Key Relations

Column	References	Notes
`admission_id`	`inpatient_admissions(id)`	CASCADE DELETE
`encounter_id`	`encounters(id)`	

8. Surgery (Operasi / OK)

Manages the full surgical workflow from surgery request to post-operative recovery. Tracks pre-operative assessment, clearance, scheduling in the OK (Kamar Operasi), intra-operative recording, and PACU (recovery room) notes.

surgery_requests

FHIR: FHIR Procedure (is_surgery=TRUE)

Purpose: Represents a surgery from initial request through completion. Can originate from outpatient, inpatient, or IGD encounters. Tracks surgery status through SURGERY_REQUESTED → SURGERY_SCHEDULED → READY_FOR_SURGERY → INTRA_OPERATIVE → SURGERY_COMPLETED → POST_OPERATIVE.

Fields

`id`	`UUID`	Primary key	NO	PK
`patient_id`	`UUID`	Patient	NO	FK → patients
`encounter_id`	`UUID`	Source encounter (requesting)	NO	FK → encounters

`episode_of_care_id`	`UUID`	Linked inpatient episode if any	YES	FK → episodes_of_care
`requested_by`	`UUID`	Doctor requesting surgery	NO	FK → practitioners
`surgery_type`	`TEXT`	Type/name of surgery	NO	
`indication`	`TEXT`	Clinical indication for surgery	NO	
`anesthesia_type`	`TEXT`	umum, lokal, regional, spinal	YES	
`status`	`surgery_status`	Current surgery workflow status	NO	ENUM
`scheduled_date`	`TIMESTAMPTZ`	Scheduled surgery date & time	YES	
`ok_location_id`	`UUID`	Operating room	YES	FK → locations
`surgeon_id`	`UUID`	Primary surgeon	YES	FK → practitioners
`anesthesiologist_id`	`UUID`	Anesthesiologist	YES	FK → practitioners
`pre_op_assessment`	`TEXT`	Pre-operative assessment notes	YES	
`surgery_start_at`	`TIMESTAMPTZ`	Actual start time	YES	
`surgery_end_at`	`TIMESTAMPTZ`	Actual end time	YES	
`intra_op_notes`	`TEXT`	Intra-operative notes	YES	
`post_op_notes`	`TEXT`	Post-operative notes	YES	
`pacu_notes`	`TEXT`	Recovery room (PACU) notes	YES	
`needs_inpatient_after`	`BOOLEAN`	Requires inpatient admission post-op	NO	DEFAULT FALSE

Foreign Key Relations

Column	References	Notes
`patient_id`	`patients(id)`	
`encounter_id`	`encounters(id)`	
`surgeon_id`	`practitioners(id)`	
`anesthesiologist_id`	`practitioners(id)`	
`ok_location_id`	`locations(id)`	

9. Laboratory

Manages the lab workflow from doctor order to results. Uses LOINC codes for standardized test identification. Integrates with Satu Sehat via FHIR ServiceRequest, Specimen, and DiagnosticReport.

lab_orders

FHIR: FHIR ServiceRequest + DiagnosticReport

Purpose: Created by a doctor during any encounter (outpatient, inpatient, IGD) when lab tests are needed. The lab nurse picks up the order, collects the specimen, processes it, and uploads results. Status flows from LAB_ORDERED → SAMPLE_TAKEN → PROCESSING → RESULT_UPLOADED.

Fields

`id`	`UUID`	Primary key	NO	PK
`encounter_id`	`UUID`	Parent encounter	NO	FK → encounters
`patient_id`	`UUID`	Patient	NO	FK → patients
`ordered_by`	`UUID`	Doctor who ordered	NO	FK → practitioners
`lab_nurse_id`	`UUID`	Lab nurse assigned	YES	FK → practitioners
`order_date`	`TIMESTAMPTZ`	When the order was placed	NO	
`status`	`lab_order_status`	Workflow status	NO	ENUM
`priority`	`TEXT`	routine, urgent, stat	NO	DEFAULT routine
`ss_service_request_id`	`TEXT`	Satu Sehat ServiceRequest ID	YES	UNIQUE
`ss_diagnostic_report_id`	`TEXT`	Satu Sehat DiagnosticReport ID	YES	UNIQUE

Foreign Key Relations

Column	References	Notes
`encounter_id`	`encounters(id)`	CASCADE DELETE
`ordered_by`	`practitioners(id)`	
`lab_nurse_id`	`practitioners(id)`	

lab_order_items

FHIR: FHIR Observation (lab) + Specimen

Purpose: Individual test items within a lab order (e.g. CBC, blood glucose, urinalysis). Each item has its own LOINC code, specimen details, and result fields. Results are entered by the lab nurse after processing.

Fields

`id`	`UUID`	Primary key	NO	PK
`lab_order_id`	`UUID`	Parent lab order	NO	FK → lab_orders
`loinc_code`	`TEXT`	LOINC code for the test	NO	
`test_name`	`TEXT`	Human-readable test name	NO	
`specimen_type`	`TEXT`	darah, urine, feses, sputum, etc.	YES	
`specimen_collected_at`	`TIMESTAMPTZ`	When specimen was collected	YES	
`ss_specimen_id`	`TEXT`	Satu Sehat Specimen ID	YES	UNIQUE
`result_value`	`TEXT`	Result value	YES	
`result_unit`	`TEXT`	Unit of measurement	YES	
`reference_range`	`TEXT`	Normal reference range	YES	
`result_status`	`TEXT`	normal, abnormal, critical	YES	
`result_entered_at`	`TIMESTAMPTZ`	When result was entered	YES	

`result_entered_by`	`UUID`	Lab nurse who entered results	YES	FK → practitioners

Foreign Key Relations

Column	References	Notes
`lab_order_id`	`lab_orders(id)`	CASCADE DELETE
`result_entered_by`	`practitioners(id)`	

10. Nutrition (Gizi)

The nutritionist's module. Only active during inpatient care. The nutritionist assesses each patient's dietary needs based on their diagnosis and condition, then creates and updates a daily meal plan throughout the stay.

nutrition_orders

FHIR: FHIR NutritionOrder

Purpose: Created by the nutritionist for inpatient patients. Contains nutritional assessment, daily caloric and protein requirements, and a flexible JSONB meal_plan (breakfast, lunch, dinner). Updated daily as the patient's condition changes. Sent to Satu Sehat as FHIR NutritionOrder.

Fields

`id`	`UUID`	Primary key	NO	PK
`episode_of_care_id`	`UUID`	Parent inpatient episode	NO	FK → episodes_of_care
`patient_id`	`UUID`	Patient	NO	FK → patients
`nutritionist_id`	`UUID`	Nutritionist managing this order	NO	FK → practitioners
`encounter_id`	`UUID`	Encounter when order was made	YES	FK → encounters
`order_date`	`DATE`	Date of this nutrition plan	NO	DEFAULT CURRENT_DATE
`nutritional_status`	`TEXT`	baik, kurang, lebih, buruk	YES	
`dietary_restrictions`	`TEXT`	e.g. diabetes diet, low sodium	YES	
`energy_needs_kcal`	`INT`	Daily caloric needs in kcal	YES	
`protein_needs_g`	`DECIMAL(6,2)`	Daily protein needs in grams	YES	
`meal_plan`	`JSONB`	JSON meal plan per meal time	YES	e.g. {pagi:...,siang:...,malam:...}
`is_active`	`BOOLEAN`	Currently active plan	NO	DEFAULT TRUE
`ss_nutrition_order_id`	`TEXT`	Satu Sehat NutritionOrder ID	YES	UNIQUE

Foreign Key Relations

Column	References	Notes
`episode_of_care_id`	`episodes_of_care(id)`	CASCADE DELETE
`nutritionist_id`	`practitioners(id)`	

11. Pharmacy

Complete pharmacy management covering the drug master (synced with Satu Sehat KFA), stock inventory, prescription management, and medication dispensing. Stock is automatically deducted when medication is dispensed via a database trigger.

medications

FHIR: FHIR Medication + KFA

Purpose: Master list of all medications available at the facility. Synced with Satu Sehat's KFA (Kamus Farmasi dan Alat Kesehatan) catalog. Contains drug classification, form, strength, and flags for narcotics/psychotropics.

Fields

`id`	`UUID`	Primary key	NO	PK
`organization_id`	`UUID`	Owning organization	NO	FK → organizations
`kfa_code`	`TEXT`	KFA product code from Satu Sehat	YES	UNIQUE
`name`	`TEXT`	Medication name	NO	
`generic_name`	`TEXT`	Generic (INN) name	YES	
`form`	`TEXT`	tablet, kapsul, sirup, injeksi, etc.	YES	
`strength`	`TEXT`	e.g. 500mg, 250mg/5ml	YES	
`unit`	`TEXT`	Dispensing unit: tablet, ml, vial	NO	DEFAULT tablet
`is_narcotics`	`BOOLEAN`	Narcotic drug flag	NO	DEFAULT FALSE
`is_psychotropics`	`BOOLEAN`	Psychotropic drug flag	NO	DEFAULT FALSE
`requires_prescription`	`BOOLEAN`	Requires doctor prescription	NO	DEFAULT TRUE
`ss_medication_id`	`TEXT`	Satu Sehat Medication ID	YES	UNIQUE

Foreign Key Relations

Column	References	Notes
`organization_id`	`organizations(id)`	

medication_stock

FHIR: — (internal inventory)

Purpose: Tracks inventory levels per medication batch. Supports multiple batches with different expiry dates. The is_below_minimum computed column is used by the pharmacist's stock monitoring dashboard to trigger purchase orders. Stock is auto-deducted by trigger when dispensed.

Fields

id	UUID	Primary key	NO	PK
medication_id	UUID	The medication	NO	FK → medications
organization_id	UUID	Owning organization	NO	FK → organizations
batch_number	TEXT	Manufacturer batch number	YES	
expiry_date	DATE	Expiry date of this batch	YES	
quantity	INT	Current stock quantity	NO	DEFAULT 0
minimum_stock	INT	Minimum stock threshold for alerts	NO	DEFAULT 10
unit_price	DECIMAL(12,2)	Purchase price per unit	NO	
is_below_minimum	BOOLEAN	COMPUTED: quantity <= minimum_stock	—	GENERATED STORED

Foreign Key Relations

Column	References	Notes
medication_id	medications(id)	CASCADE DELETE
organization_id	organizations(id)	

prescriptions

FHIR: FHIR MedicationRequest

Purpose: Header record for a prescription written by a doctor during an encounter. A prescription can contain multiple items (prescription_items). Sent to Satu Sehat as FHIR MedicationRequest.

Fields

id	UUID	Primary key	NO	PK
encounter_id	UUID	Source encounter	NO	FK → encounters
patient_id	UUID	Patient	NO	FK → patients
prescribed_by	UUID	Prescribing doctor	NO	FK → practitioners
prescription_date	TIMESTAMPTZ	Date of prescription	NO	
status	TEXT	active, completed, cancelled	NO	
ss_medication_request_id	TEXT	Satu Sehat MedicationRequest ID	YES	UNIQUE

Foreign Key Relations

Column	References	Notes
encounter_id	encounters(id)	CASCADE DELETE
prescribed_by	practitioners(id)	

prescription_items

FHIR: — (part of MedicationRequest)

Purpose: Individual drug items within a prescription. Each item specifies dosage, frequency, duration, and quantity. The pharmacist marks is_dispensed = TRUE after dispensing.

Fields

`id`	`UUID`	Primary key	NO	PK
`prescription_id`	`UUID`	Parent prescription	NO	FK → prescriptions
`medication_id`	`UUID`	The medication prescribed	NO	FK → medications
`dosage`	`TEXT`	Dosage instruction e.g. 3x1 tablet	NO	
`frequency`	`TEXT`	Frequency e.g. setiap 8 jam	YES	
`duration_days`	`INT`	Treatment duration in days	YES	
`quantity`	`INT`	Total units to dispense	NO	
`instructions`	`TEXT`	Patient instructions e.g. after meal	YES	
`is_dispensed`	`BOOLEAN`	Marked TRUE after pharmacist dispenses	NO	DEFAULT FALSE

Foreign Key Relations

Column	References	Notes
`prescription_id`	`prescriptions(id)`	CASCADE DELETE
`medication_id`	`medications(id)`	

medication_dispenses

FHIR: FHIR MedicationDispense

Purpose: Records actual medication dispensing by the pharmacist. Linked to a specific stock batch. A database trigger automatically deducts the quantity from medication_stock when a new dispense record is inserted. Sent to Satu Sehat as FHIR MedicationDispense.

Fields

`id`	`UUID`	Primary key	NO	PK
`prescription_id`	`UUID`	Parent prescription	NO	FK → prescriptions
`encounter_id`	`UUID`	Source encounter	NO	FK → encounters
`patient_id`	`UUID`	Patient	NO	FK → patients
`dispensed_by`	`UUID`	Pharmacist who dispensed	NO	FK → practitioners
`medication_id`	`UUID`	Medication dispensed	NO	FK → medications
`stock_id`	`UUID`	Specific stock batch used	YES	FK → medication_stock
`quantity_dispensed`	`INT`	Quantity given to patient	NO	
`dispensed_at`	`TIMESTAMPTZ`	Dispense timestamp	NO	

ss_medication_dispense_id	TEXT	Satu Sehat MedicationDispense ID	YES	UNIQUE

Foreign Key Relations

Column	References	Notes
prescription_id	prescriptions(id)	
dispensed_by	practitioners(id)	
medication_id	medications(id)	
stock_id	medication_stock(id)	Trigger deducts quantity on INSERT

12. Purchase Orders

Pharmacy procurement workflow. The pharmacist monitors stock levels (is_below_minimum flag), creates PO drafts, which are reviewed and approved by management, then sent to vendors. On receipt, stock is automatically added via trigger.

vendors

FHIR: — (internal)

Purpose: Master list of pharmaceutical suppliers. Each vendor is linked to an organization. Used in purchase orders to select the supplier for each procurement.

Fields

id	UUID	Primary key	NO	PK
organization_id	UUID	Owning organization	NO	FK → organizations
name	TEXT	Vendor company name	NO	
contact_person	TEXT	Contact person name	YES	
phone	TEXT	Contact phone	YES	
email	TEXT	Contact email	YES	
address	TEXT	Vendor address	YES	
is_active	BOOLEAN	Active status	NO	DEFAULT TRUE

Foreign Key Relations

Column	References	Notes
organization_id	organizations(id)	

purchase_orders

FHIR: — (internal)

Purpose: Purchase order header. Tracks the full procurement lifecycle: PO_DRAFT → management approval → PO_SENT → goods received → PO_COMPLETED. Requires management approval before being sent to vendors.

Fields

id	UUID	Primary key	NO	PK
organization_id	UUID	Owning organization	NO	FK → organizations
vendor_id	UUID	Target vendor	NO	FK → vendors
po_number	TEXT	Unique PO number	NO	UNIQUE
order_date	DATE	Date PO was created	NO	
expected_delivery_date	DATE	Expected goods arrival date	YES	
status	po_status	PO workflow status	NO	ENUM
total_amount	DECIMAL	Total PO value	NO	DEFAULT 0
created_by	UUID	Pharmacist who created PO	NO	FK → practitioners
approved_by	UUID	Manager who approved	YES	FK → practitioners
approved_at	TIMESTAMPTZ	Approval timestamp	YES	
rejection_reason	TEXT	Reason if PO was rejected	YES	

Foreign Key Relations

Column	References	Notes
organization_id	organizations(id)	
vendor_id	vendors(id)	
created_by	practitioners(id)	
approved_by	practitioners(id)	

purchase_order_items

FHIR: — (internal)

Purpose: Line items within a purchase order. Each item tracks ordered quantity, unit price (subtotal auto-calculated), received quantity, expiry date, and batch number. A trigger automatically adds received stock to medication_stock when quantity_received is updated.

Fields

id	UUID	Primary key	NO	PK
po_id	UUID	Parent purchase order	NO	FK → purchase_orders
medication_id	UUID	Medication being ordered	NO	FK → medications
quantity_ordered	INT	Quantity ordered from vendor	NO	

unit_price	DECIMAL	Price per unit	NO	
subtotal	DECIMAL	COMPUTED: quantity_ordered * unit_price	—	GENERATED STORED
quantity_received	INT	Quantity actually received	NO	DEFAULT 0
received_date	DATE	Date goods were received	YES	
expiry_date	DATE	Expiry date of received batch	YES	
batch_number	TEXT	Batch number from manufacturer	YES	
is_fully_received	BOOLEAN	COMPUTED: quantity_received >= ordered	—	GENERATED STORED

Foreign Key Relations

Column	References	Notes
po_id	purchase_orders(id)	CASCADE DELETE
medication_id	medications(id)	

13. Billing & Invoicing

Handles patient billing for all service types. Outpatient patients pay at the cashier after the encounter. Inpatient charges accumulate in the running_bill table and are converted to a final invoice at discharge. BPJS claims are tracked separately.

invoices

FHIR: FHIR Invoice

Purpose: Final invoice for a patient encounter. For outpatient and IGD, created at the cashier after the visit. For inpatient, created at discharge by consolidating the running_bill. BPJS payments are tracked via claim fields. Sent to Satu Sehat as FHIR Invoice.

Fields

id	UUID	Primary key	NO	PK
encounter_id	UUID	Source encounter	NO	FK → encounters
patient_id	UUID	Patient	NO	FK → patients
organization_id	UUID	Organization	NO	FK → organizations
invoice_number	TEXT	Unique invoice number (auto-gen)	NO	UNIQUE
invoice_date	TIMESTAMPTZ	Invoice creation timestamp	NO	
payment_type	payment_type	umum or bpjs	NO	ENUM
subtotal	DECIMAL	Pre-discount total	NO	
discount_amount	DECIMAL	Total discount applied	NO	
total_amount	DECIMAL	Final amount payable	NO	

paid_amount	DECIMAL	Amount actually paid	NO	
status	TEXT	unpaid, paid, bpjs_claim_pending...	NO	
paid_at	TIMESTAMPTZ	Payment timestamp	YES	
cashier_id	UUID	Cashier who processed payment	YES	FK → practitioners
bpjs_sep_number	TEXT	BPJS SEP number for claim	YES	
ss_invoice_id	TEXT	Satu Sehat Invoice ID	YES	UNIQUE

Foreign Key Relations

Column	References	Notes
encounter_id	encounters(id)	
organization_id	organizations(id)	
cashier_id	practitioners(id)	

invoice_items

FHIR: FHIR ChargeItem

Purpose: Line items on an invoice. Each item can be a consultation fee, medication cost, lab fee, tindakan fee, room charge, or nutrition charge. The reference_id links to the specific record that generated the charge.

Fields

id	UUID	Primary key	NO	PK
invoice_id	UUID	Parent invoice	NO	FK → invoices
item_type	TEXT	consultation, medication, lab, action, room, nutrition	NO	
item_name	TEXT	Description of the charge	NO	
item_code	TEXT	Service/item code	YES	
quantity	INT	Quantity of the item	NO	DEFAULT 1
unit_price	DECIMAL	Price per unit	NO	
subtotal	DECIMAL	COMPUTED: quantity * unit_price	—	GENERATED STORED
reference_id	UUID	ID of source record (procedure, dispense, etc.)	YES	

Foreign Key Relations

Column	References	Notes
invoice_id	invoices(id)	CASCADE DELETE

running_bills

FHIR: — (internal pre-invoice)

Purpose: Accumulates charges for inpatient patients on a daily basis before the final invoice is generated at discharge. Each room charge, medication, lab test, and tindakan is added here automatically. At discharge, this table is summarized into the final invoice.

Fields

`id`	`UUID`	Primary key	NO	PK
`episode_of_care_id`	`UUID`	Parent inpatient episode	NO	FK → episodes_of_care
`patient_id`	`UUID`	Patient	NO	FK → patients
`item_type`	`TEXT`	room, medication, lab, action, nutrition	NO	
`item_name`	`TEXT`	Charge description	NO	
`quantity`	`INT`	Quantity	NO	DEFAULT 1
`unit_price`	`DECIMAL`	Price per unit	NO	
`subtotal`	`DECIMAL`	COMPUTED: quantity * unit_price	—	GENERATED STORED
`charge_date`	`DATE`	Date the charge was incurred	NO	DEFAULT CURRENT_DATE
`reference_id`	`UUID`	Source record ID	YES	

Foreign Key Relations

Column	References	Notes
`episode_of_care_id`	`episodes_of_care(id)`	CASCADE DELETE

14. System & Integration Tables

Infrastructure tables for Satu Sehat API synchronization tracking and full audit logging.

ss_sync_logs

FHIR: — (integration log)

Purpose: Every API call to Satu Sehat is logged here. Records the FHIR resource type, local record ID, action (POST/PUT/PATCH), full request and response payloads, HTTP status, and retry count. Used by admin to monitor integration health and debug sync failures.

Fields

`id`	`UUID`	Primary key	NO	PK
`resource_type`	`TEXT`	FHIR resource e.g. Encounter	NO	
`local_id`	`UUID`	ID in our local database	NO	
`ss_resource_id`	`TEXT`	UUID returned by Satu Sehat	YES	
`action`	`TEXT`	HTTP method: POST, PUT, PATCH	NO	

request_payload	JSONB	Full FHIR payload sent	YES	
response_payload	JSONB	Full response from Satu Sehat	YES	
http_status	INT	HTTP status code received	YES	
status	TEXT	success or failed	NO	
error_message	TEXT	Error detail if failed	YES	
retry_count	INT	Number of retry attempts	NO	DEFAULT 0
synced_at	TIMESTAMPTZ	Timestamp of sync attempt	NO	

audit_logs

FHIR: — (compliance/audit)

Purpose: Complete audit trail for all data changes and user actions in the system. Records who changed what data, when, and from where. Required for medical record compliance. Old and new values are stored as JSONB for full change history.

Fields

id	UUID	Primary key	NO	PK
user_id	UUID	User who performed the action	YES	FK → auth.users
action	TEXT	INSERT, UPDATE, DELETE, LOGIN, ACCESS	NO	
table_name	TEXT	Name of the table affected	YES	
record_id	UUID	ID of the affected record	YES	
old_data	JSONB	Previous values before change	YES	
new_data	JSONB	New values after change	YES	
ip_address	INET	Client IP address	YES	
user_agent	TEXT	Browser/client user agent	YES	
created_at	TIMESTAMPTZ	Timestamp of the action	NO	

Foreign Key Relations

Column	References	Notes
user_id	auth.users(id)	